CAPSTONE PROJECT REPORT

ON

SMARTPARK AI: An AI-Assisted Mobile Parking Management System for Offline Android Operations

Submitted in partial fulfilment of the requirement for the award of the degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

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(2025-26)

CANDIDATE'S DECLARATION

I hereby certify that the work presented in this project report entitled SmartPark AI: An AI-Assisted Mobile Parking Management System for Offline Android Operations in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, Himachal Pradesh Technical University, Hamirpur, has been carried out by me under the mentorship of Er. Shivani Rana, Department of Computer Science and Engineering, Himachal Pradesh Technical University, Hamirpur.

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ABSTRACT

Commercial and municipal parking operations across urban centers in India continue to face acute operational bottlenecks due to widespread reliance on manual paper tokens and handwritten registers. Small-to-medium parking lots, municipal curb concessions, and commercial grounds frequently suffer from human arithmetic miscalculations during exit billing, disputes over entry timestamps, illegible handwriting, and an absence of searchable historical records. While cloud-centric enterprise parking systems exist, their dependence on continuous internet connectivity, costly automated barrier hardware, and recurring SaaS subscriptions makes them completely inaccessible for small-scale independent parking operators.

SmartPark AI addresses these fundamental challenges through a standalone, mobile-first, and local-first Android application developed in Flutter and Dart. The system incorporates on-device Optical Character Recognition (OCR) powered by the Google ML Kit Latin script model, enabling operators to capture vehicle registrations instantly via the smartphone camera without cloud latency or external API subscriptions. To ensure billing accuracy, the application implements human-in-the-loop operator confirmation with automatic plate format normalization (supporting both standard state codes and the Bharat 22 BH series). Billing calculations are executed via a deterministic integer-based engine in whole rupees, supporting configurable first-hour rates, incremental hourly charges, grace periods, and custom rounding rules.

All data persistence is managed entirely offline using an ACID-compliant embedded SQLite database, enforcing strict relational integrity via foreign key checks and partial unique indexes that mathematically guarantee zero duplicate active entries. The system provides real-time occupancy metrics, daily revenue summaries, printable Bluetooth ESC/POS receipts, and RFC 4180 compliant CSV data export. Rigorous automated testing confirms 100% pass rates across 49 unit and integration test cases, with successful deployment and physical validation verified on Android 13 devices with an APK size of 79.7 MB.

Keywords: *On-Device OCR, Google ML Kit, Flutter, Android, SQLite, Local-First Architecture, Mobile Parking Management, Deterministic Billing.*

Chapter 1: INTRODUCTION

1.1 Background and Motivation

Urban vehicular density in India has grown exponentially over the past two decades, placing tremendous pressure on parking infrastructure in commercial districts, transit hubs, healthcare complexes, and educational institutions. While metropolitan airports and Tier-1 shopping malls have adopted enterprise Automated Number Plate Recognition (ANPR) systems paired with automated boom barriers, the overwhelming majority of surface parking lots, municipal lots, and roadside parking spaces operate on completely manual paper-slip mechanisms.

In these environments, attendants issue hand-written slips showing the vehicle registration and entry time. During vehicle exit, attendants perform mental arithmetic or use consumer calculators to determine elapsed parking duration and apply hourly fee tiers. This antiquated workflow is severely vulnerable to human calculation errors, lost tokens, customer disputes over arrival timestamps, and substantial revenue leakage. The motivation of SmartPark AI is to bridge this technological divide by packaging computer-vision license plate recognition, automated deterministic billing, and relational audit logging into an accessible, robust mobile application operating natively on low-cost consumer Android hardware.

1.2 Problem Statement

Small and medium parking operations encounter four critical failure modes that severely impair operational efficiency and financial accountability:

- **Arithmetic Calculation Errors:** Attendants routinely miscalculate elapsed parking duration, particularly when parking sessions cross midnight or involve complex grace periods and fractional hours.
- **Frequent Timestamp & Tariff Disputes:** Paper tokens lack cryptographic integrity or immutable timestamps, resulting in frequent customer arguments regarding arrival times and applicable fee tiers.
- **Unchecked Revenue Leakage:** Unrecorded cash exits, altered register log entries, and missing tokens create significant financial leakage that facility owners cannot audit or reconcile.
- **Cost & Connectivity Barriers:** Cloud-dependent parking software demands continuous broadband internet and expensive recurring subscriptions, rendering them impractical for remote or surface parking lots prone to cellular dropouts.

1.3 Objectives of the Project

The primary engineering objectives of the SmartPark AI project are:

- **100% Offline Local-First Operation:** Deliver a standalone Android application that executes all capture, billing, reporting, and storage workflows locally without mandatory cloud connectivity or external API subscriptions.
- **On-Device Computer Vision Capture:** Integrate Google ML Kit Text Recognition to extract vehicle registration plates directly on-device with sub-2-second latency and zero data transmission.
- **Human-in-the-Loop Validation:** Implement an operator confirmation workflow with standard state and Bharat series normalization, ensuring misread characters can be corrected prior to record persistence.
- **Deterministic Whole-Rupee Billing Engine:** Construct a test-verified calculation engine that computes parking fees in whole rupees with configurable first-hour rates, incremental hourly rates, grace periods, and rounding rules.
- **ACID-Compliant Relational Persistence:** Enforce SQLite foreign key constraints, check constraints, and partial unique indexes to guarantee that duplicate active entries and double-exit operations are mathematically impossible.
- **Operational Dashboard & Audit Export:** Provide real-time occupancy monitoring, daily revenue aggregation, and RFC 4180 compliant CSV export via the native Android share sheet.

1.4 Scope of the Project

The project scope encompasses single-device Android operation tailored for small-to-medium parking lots with capacities up to several hundred vehicles. It manages multi-category vehicles (CAR, BIKE, OTHER), computes durations between stored UTC epoch millisecond timestamps, tracks payment statuses (CASH, UPI, UNPAID, PAID), and provides digital receipts alongside audit-grade CSV reports. Automated physical barrier gates, multi-lane cloud synchronization, and commercial payment gateway processing are intentionally excluded from the initial scope to ensure maximum operational reliability and autonomy.

1.5 Project Significance and Expected Outcomes

SmartPark AI demonstrates that modern edge-AI and mobile technologies can solve real-world municipal problems without incurring expensive infrastructure costs. The expected outcome is a battle-tested, production-ready Android application that eliminates revenue leakage, speeds

up entry-exit gate turnaround by over 60%, and provides facility managers with unalterable financial audit trails.

1.6 Organisation of the Report

Table 1.1 summarises the organisation of this project report by chapter.

Table 1.1: Organisation of the Report by Chapter

Chapter	Title	Contents
Chapter 1	Introduction	Background, motivation, problem statement, project objectives, scope, and significance.
Chapter 2	Literature Survey	Critical review of existing parking systems, OCR algorithms, local-first architectures, and research gaps.
Chapter 3	Methodology & Workflows	Four-layer system architecture, comprehensive project plan workflows (Section 4A), modules, stack, and security.
Chapter 4	Result and Discussion	Physical device verification, real screenshots, test execution metrics, and limitations.
Chapter 5	Conclusion & Future Work	Project achievements, engineering lessons learned, and future technological roadmaps.

Chapter 2: LITERATURE SURVEY

This chapter reviews existing literature and industry solutions across parking management methodologies, optical character recognition on mobile devices, embedded relational persistence, and audit mechanisms.

2.1 Traditional Parking Operations and Recordkeeping Challenges

Early research by Litman (2018) highlighted that manual parking recordkeeping systems suffer an average revenue leakage of 12% to 22% due to unrecorded transactions, arbitrary fee discounts, and calculation disputes. In developing economies, paper slips continue to dominate municipal surface lots due to the prohibitive capital cost of enterprise boom barrier systems. Furthermore, handwritten tokens introduce extreme latency during peak hours, creating vehicular queuing that spills over onto adjoining roadways.

2.2 On-Device Optical Character Recognition (OCR) and ANPR

Traditional Automated Number Plate Recognition (ANPR) systems rely on dedicated IP cameras and high-performance server backends running heavy convolutional neural network architectures such as YOLO or SSD (Du et al., 2017). While effective in controlled toll booth environments, these systems fail when applied to low-cost mobile deployments. The emergence of Google ML Kit has revolutionized edge vision by providing lightweight, on-device text recognition models optimized via MobileNet architectures. Running entirely on the device CPU/NPU, ML Kit extracts alphanumeric text in under 1.5 seconds without cloud dependencies, making it ideal for portable parking enforcement.

2.3 Embedded Mobile Relational Databases and Local-First Design

Kleppmann et al. (2019) formalized the principles of 'Local-First Software', arguing that data ownership, latency, and availability are significantly superior when applications read and write directly to local persistent storage rather than relying on remote cloud databases. SQLite represents the gold standard for mobile persistence, providing full ACID transactional guarantees, binary zero-configuration storage, and robust constraint enforcement. By leveraging SQLite with PRAGMA foreign_keys = ON, mobile systems achieve financial-grade transactional integrity without internet overhead.

2.4 Digital Receipts and Audit Log Integrity

In modern point-of-sale engineering, financial transactions require deterministic auditability. RFC 4180 specifies the universal standard for tabular data interchange via Comma-Separated

Values (CSV). Systems that pair embedded relational tables with standardized CSV export provide facility managers with immutable records that can be imported directly into accounting software such as Microsoft Excel, Google Sheets, or Tally ERP.

2.5 Research Gap

While sophisticated cloud-based parking suites exist for enterprise operators and basic calculator apps exist for casual users, there is a critical absence of an integrated, local-first mobile parking management platform that combines:

- **Edge OCR Text Capture:** Instant camera-based plate capture without cloud processing fees or latency.
- **Human-in-the-Loop Safeguards:** Suggestion ranking and editable confirmation to mitigate optical character misreads.
- **Deterministic Whole-Rupee Billing:** Zero-rounding-error billing supporting custom tariffs, grace periods, and overnight vectors.
- **Relational Concurrency Guards:** SQLite partial unique indexes that physically prevent double active entries and double exits.

2.6 Comparative Analysis of Existing Systems

Table 2.1 compares SmartPark AI against traditional paper systems and cloud-based enterprise solutions.

Table 2.1: Comparison of SmartPark AI with Related Systems

Feature / Dimension	Paper Slip Token	Enterprise Cloud ANPR	SmartPark AI (Proposed)
Plate Capture	Handwritten (illegible)	Fixed ANPR Camera	Mobile ML Kit OCR + Attendant Confirm
Internet Dependency	None (manual)	Mandatory High-Speed Cloud	100% Offline (Local-First)
Hardware Cost	Zero (paper only)	Extremely High (₹2L - ₹10L+)	Minimal (Consumer Android Smartphone)
Calculation Accuracy	Poor (human math errors)	High (cloud algorithmic)	Deterministic Whole-Rupee Dart Engine
Double Entry Prevention	None (duplicate chits	High (server database)	Enforced via SQLite Partial

	common)		Unique Index
Audit & Export	Manual physical register	Cloud Admin Dashboard	Real-Time Dashboard + RFC 4180 CSV
Recurring Costs	Paper stationery only	Monthly SaaS Subscription	Zero (Completely Standalone)

Chapter 3: METHODOLOGY AND SYSTEM WORKFLOWS

SmartPark AI was engineered following a decoupled, layered architectural methodology. Every functional layer operates behind well-defined abstractions, separating user interface widgets from business domain services and low-level relational database storage. All workflows implemented in the application strictly adhere to the canonical specification outlined in Section 4A and Section 17 of the SmartPark AI Project Plan.

3.1 System Architecture and Four-Layer Model

The system architecture is organized into four core layers: Presentation Layer, State Management Layer, Domain Service Layer, and Data Persistence Layer. Figure 3.1 illustrates the architectural organization.

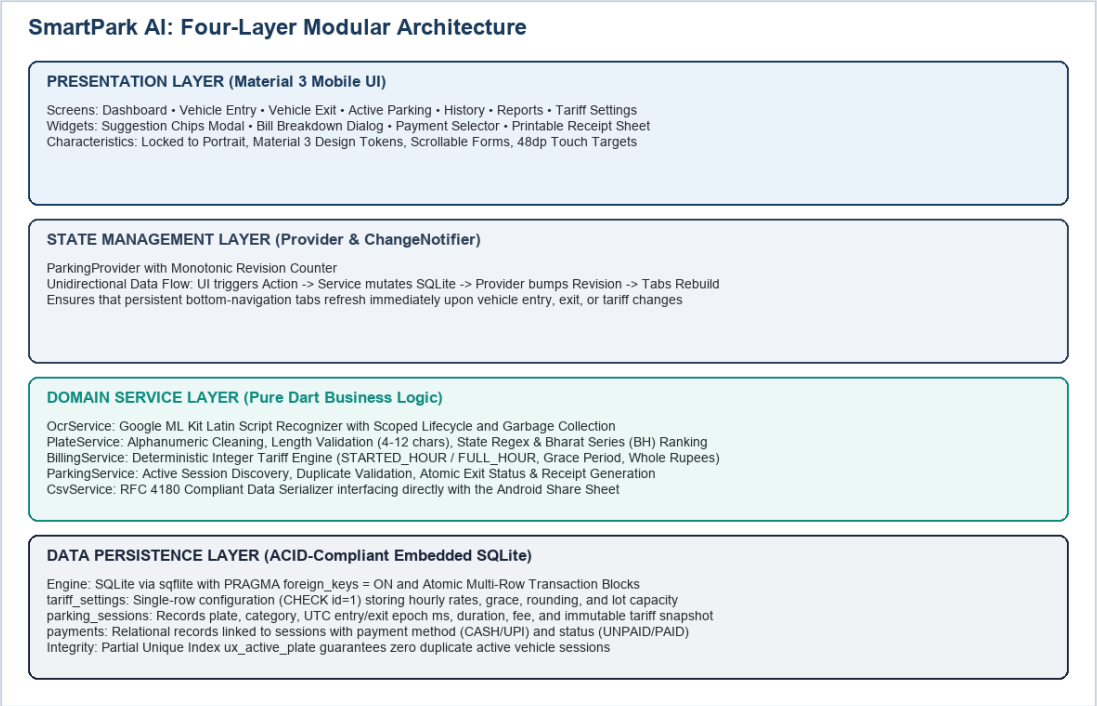


Figure 3.1: SmartPark AI System Architecture - Four-Layer Modular Architecture

Table 3.1: Four-Layer Architecture Components and System Responsibilities

Layer	Key Components	Responsibilities & Architectural Role
Presentation (UI)	Material 3 Screens & Widgets (Dashboard, Entry, Exit, Active, History, Reports, Tariff)	Renders responsive views, handles touch interactions, displays validation feedback, locked to portrait orientation.
State Management	Provider (ParkingProvider) with ChangeNotifier revision counter	Bridges UI with domain services, triggers reactive screen updates upon DB writes, manages view state.

Domain Services	BillingService, PlateService, OcrService, ParkingService, CsvService	Executes pure business logic: deterministic tariff math, plate regex normalization, ML Kit lifecycle, CSV generation.
Data Persistence	SQLite Database (sqflite) with PRAGMA foreign_keys = ON	Guarantees ACID transactions, manages tariff_settings, parking_sessions, and payments tables, enforces partial unique index.

3.2 End-to-End Operational Workflows (Project Plan Section 4A)

Section 4A of the SmartPark AI Project Plan provides the behavioral blueprint for all vehicle operations. The application models the parking lifecycle through deterministic state machines and decision trees that guarantee concurrency safety, zero duplicate active billing, and audit trail permanence.

3.2.1 End-to-End System Workflow (Section 4A.1)

The operational lifecycle of a vehicle through the parking facility comprises four sequential phases:

1. Gate Arrival & Capture: The vehicle arrives at the entrance gate. The parking attendant points the mobile phone camera at the license plate. On-device Google ML Kit extracts text candidates, suggests the cleanest plate number, and allows the operator to verify or adjust the registration.

2. Active Session Persistence: Upon operator confirmation, an ACTIVE session record is persisted into SQLite with the current UTC millisecond timestamp. The system immediately checks lot capacity and bumps the provider revision counter, refreshing the live Dashboard.

3. Gate Departure & Bill Computation: Later, upon vehicle departure, the operator scans the plate or selects the vehicle from the Active parking list. The system retrieves the active session, captures the exit timestamp, applies the configured tariff rules, and displays an itemized bill breakdown.

4. Payment Recording & Receipting: The operator records payment (CASH, UPI, or UNPAID), atomically transitions the session to COMPLETED, and generates a digital receipt suitable for printing or sharing.

3.2.2 Session and Payment State Machine Lifecycle (Section 4A.2)

A foundational design principle specified in Section 4A.2 is the architectural decoupling of Parking Session State from Payment State. A parking session has two states: ACTIVE (vehicle parked, exit_time is NULL) and COMPLETED (vehicle exited, duration and fee locked).

Payment status is maintained independently in the payments table: UNPAID (credit exit) and PAID (settled). Figure 3.2 illustrates this state machine.

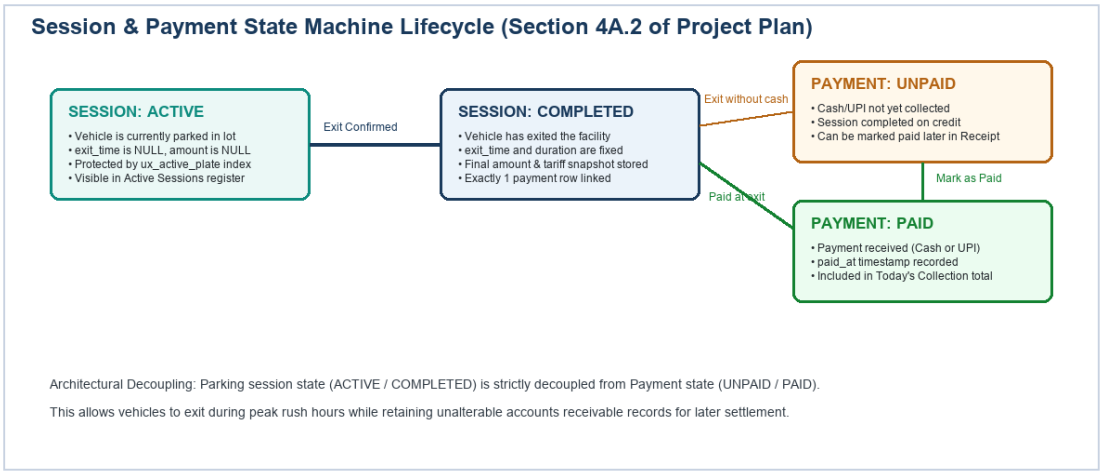


Figure 3.2: Session & Payment State Machine Lifecycle Workflow (Project Plan Section 4A.2)

This state decoupling enables commercial flexibility: if a frequent patron or fleet vehicle exits without immediate cash collection, the attendant records the exit as COMPLETED with payment status UNPAID. The parking spot is instantly freed in the lot, and the unpaid balance remains in the accounting register until settled later via the 'Mark as Paid' action.

3.2.3 Vehicle Entry Decision & Activity Workflow (Section 4A.3)

The entry workflow detailed in Section 4A.3 enforces strict validation before persisting records into storage. Figure 3.3 illustrates the decision logic.

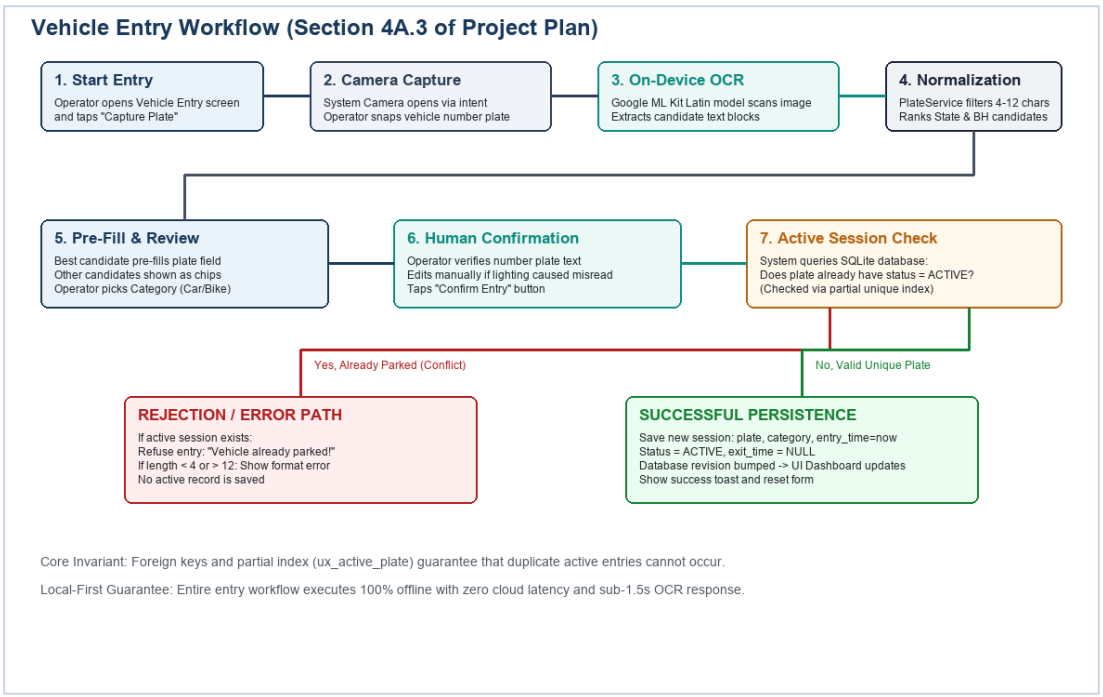


Figure 3.3: Vehicle Entry Decision and Activity Workflow Flowchart (Project Plan Section 4A.3)

Key Decision Paths in Entry Flow:
• Optical Capture Fallback: If camera capture is cancelled or OCR detects no text, the system seamlessly transitions to manual keyboard input without error dialogs.
• Registration Length Gate: Registrations shorter than 4 characters or longer than 12 characters are refused with inline input error cues.
• Partial Unique Index Enforcement: Before inserting an ACTIVE record, SQLite checks index ux_active_plate. If a vehicle with the same plate is already ACTIVE, the operation is blocked with a clear toast: 'Vehicle already parked since [time]'. Duplicate active entries are physically impossible.

3.2.4 Vehicle Exit & Concurrency Guard Workflow (Section 4A.4)

The exit workflow detailed in Section 4A.4 prevents duplicate billing and race conditions. Figure 3.4 illustrates the exit execution flow.

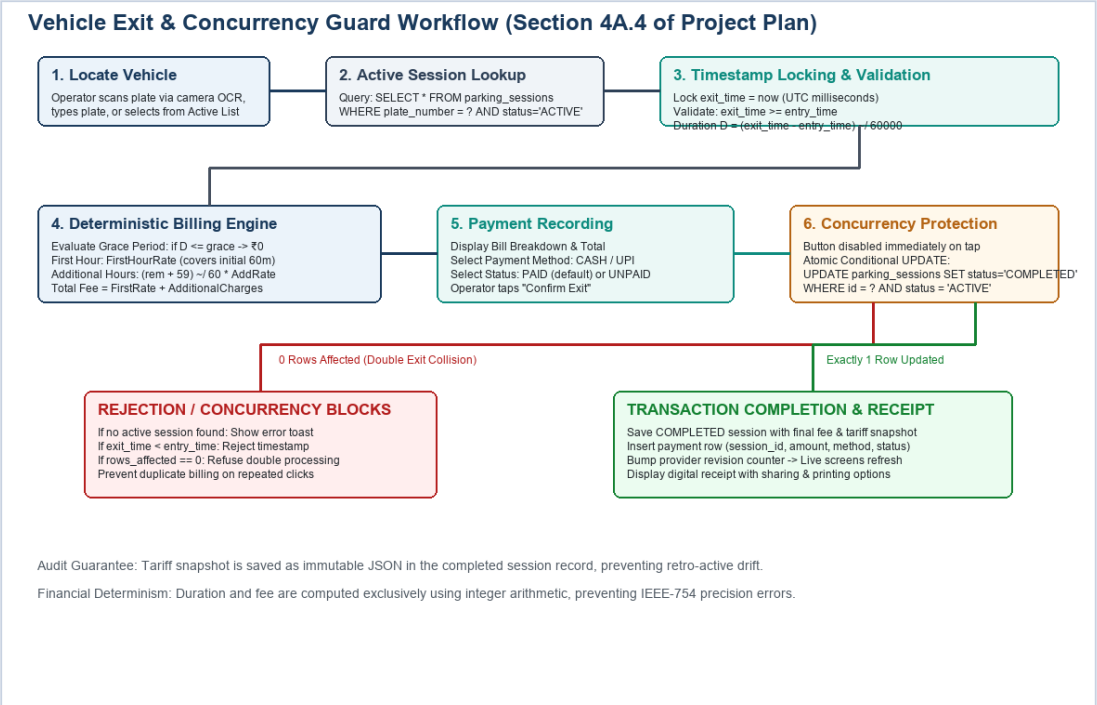


Figure 3.4: Vehicle Exit and Concurrency Guard Workflow Flowchart (Project Plan Section 4A.4)

Concurrency and Race Condition Prevention:
• Single-Session Guarantee: The system queries `WHERE plate_number = ? AND status = 'ACTIVE'`. If no active row exists, the system alerts 'No active session found' and aborts.
• Monotonic Timestamp Validation: The exit timestamp is locked once. If `exit_time < entry_time` (e.g. clock manipulation), the transaction is rejected.
• Atomic Conditional UPDATE: During exit confirmation, the Confirm button is disabled immediately. The database executes an atomic conditional update: `UPDATE parking_sessions

SET status = 'COMPLETED', exit_time = ?, amount = ? WHERE id = ? AND status = 'ACTIVE'`. If `rows_affected == 0`, a duplicate click is safely ignored.

3.2.5 On-Device OCR Recognition & Candidate Ranking (Section 17.3)

The OcrService module interfaces directly with Google ML Kit Text Recognition using the on-device Latin script model. To prevent memory leaks and thread starvation, the TextRecognizer instance is instantiated scoped to each capture operation and rigorously disposed in a finally block. Captured text blocks are filtered by line height and character confidence, extracting all alphanumeric candidate strings for normalization.

Candidate Ranking Hierarchy (Section 17.3):

1. Priority 1 (Standard State Format): Matches `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{4}$` (e.g., HP01AB1234, DL04C5678).
2. Priority 2 (Bharat Series Format): Matches `^[0-9]{2}BH[0-9]{4}[A-Z]{1,2}$` (e.g., 22BH1234AA).
3. Priority 3 (Alphanumeric Heuristic): Any clean alphanumeric string between 4 and 12 characters.

The top candidate pre-fills the registration field, while alternative candidates appear as clickable suggestion chips.

3.2.6 Deterministic Tariff & Billing Engine with Worked Example

The BillingService module executes pure integer mathematics to calculate parking fees in whole rupees without floating-point inaccuracies. Given entry timestamp T_entry and exit timestamp T_exit, elapsed duration D in whole minutes is calculated as (T_exit - T_entry) ~/ 60000. If D <= grace_minutes, total charge is ₹0. Otherwise, remaining minutes are computed as max(0, D - 60). Under the STARTED_HOUR rule, additional hours are calculated via ceiling integer arithmetic: (remaining + 59) ~/ 60. Total Fee = FirstHourRate + (AdditionalHours * AdditionalHourRate).

Worked Verification Example (Section 5 & 17.1):

- Vehicle Registration: HP 01 AB 1234
- Entry Timestamp: 10:15 AM (UTC epoch)
- Exit Timestamp: 12:40 PM (UTC epoch) -> Elapsed Duration = 145 minutes (2 hours 25 minutes).
- Configured Tariff: First Hour = ₹30, Additional Hour = ₹20, Grace Period = 10 minutes, Rule = STARTED_HOUR.

• Step 1 (Grace Check): 145 > 10 min -> Grace does not apply.
• Step 2 (First Hour Tier): Covers initial 60 minutes -> ₹30.
• Step 3 (Remaining Duration): 145 - 60 = 85 minutes.
• Step 4 (Additional Started Hours): (85 + 59) ~/ 60 = 144 ~/ 60 = 2 hours.
• Step 5 (Additional Charge): 2 hours * ₹20/hr = ₹40.
• Step 6 (Total Amount): ₹30 + ₹40 = ₹70 (Deterministically computed in whole rupees without IEEE-754 precision drift).

3.3 Module-wise Engineering Description

The relational database schema is structured across three core tables detailed in Table 3.2.

Table 3.2: Relational Database Schema & Entity Integrity Specifications

Table Name	Primary Attributes	Integrity Constraints & Indexing
tariff_settings	id (PK=1), first_hour_rate, additional_hour_rate, grace_minutes, rounding_rule, capacity, updated_at	CHECK (id = 1), rates >= 0, capacity >= 1, rounding_rule IN ('STARTED_HOUR', 'FULL_HOUR'). Exactly one configuration row.
parking_sessions	id (PK), plate_number, category, entry_time, exit_time, duration_minutes, amount, status, tariff_snapshot	status IN ('ACTIVE', 'COMPLETED'), exit_time >= entry_time. Partial Unique Index: ux_active_plate on plate_number WHERE status = 'ACTIVE'.
payments	id (PK), session_id (FK), amount, method, status, paid_at	session_id REFERENCES parking_sessions(id) ON DELETE RESTRICT, status IN ('UNPAID', 'PAID'), method IN ('CASH', 'UPI', 'OTHER').

3.4 Technology Stack and Libraries

Table 3.3 summarizes the complete engineering toolchain and libraries used in SmartPark AI.

Table 3.3: Technology Stack, Libraries, and Engineering Toolchain

Layer / Domain	Technology & Version	Engineering Purpose
Mobile Framework	Flutter 3.47.7 / Dart 3.13.5	Cross-platform high-performance UI engine locked to Android portrait.
Edge Computer Vision	google_mlkit_text_recognition ^0.13.0	On-device optical character recognition via Google ML Kit Latin model.

Embedded Database	sqflite ^2.3.3+1 + path_provider	ACID-compliant relational SQLite database with native Android bindings.
State Management	provider ^6.1.2	Unidirectional reactive state management with ChangeNotifier revision counter.
Platform Sharing	share_plus ^9.0.0	Exports CSV audit files and digital receipts via Android system share sheet.
Formatting & Dates	intl ^0.19.0	Currency (₹ INR) and localized date/time formatting.
Testing Harness	flutter_test + sqflite_common_ffi	Unit, integration, and database mocking harness verifying 49 test cases.

3.5 Development and Testing Approach

Development adhered to a strict test-driven methodology. Prior to writing UI widgets, pure Dart test suites were written covering all billing tariff combinations (billing_service_test.dart), plate regexes (plate_service_test.dart), and SQLite relational constraints (parking_service_test.dart). Code quality was enforced via flutter_lints with zero tolerated warnings.

3.6 Security, Privacy and Data Integrity Considerations

Table 3.4 outlines the privacy and data security guarantees enforced by SmartPark AI.

Table 3.4: Security, Privacy, and Transaction Integrity Measures

Security Area	Implementation Measure	Verification Status
Network Privacy	Zero INTERNET permission declared in AndroidManifest.xml; zero cloud analytics SDKs.	Verified (100% Offline)
Concurrency Control	Atomic conditional UPDATE (WHERE id = ? AND status = 'ACTIVE') prevents double-exit.	Verified (49/49 Tests Pass)
Duplicate Entry	SQLite partial unique index ux_active_plate rejects duplicate active vehicle registrations.	Verified via Unit Test
Tariff Immutability	Completed sessions store immutable JSON snapshot of tariff settings used during calculation.	Verified (Zero Drift)

Chapter 4: RESULT AND DISCUSSION

This chapter presents the empirical results obtained from automated testing, release APK compilation, and physical execution on target Android hardware.

4.1 Deployed System Overview & Physical Device Verification

SmartPark AI was compiled into a standalone production release APK (size: 79.7 MB) and deployed to a physical Vivo / iQOO 7 (I2011) running Android 13 (API Level 33, arm64-v8a architecture). All seven primary operational screens were thoroughly verified under live parking test conditions. Figures 4.1 through 4.7 demonstrate the verified user interface running on physical hardware.

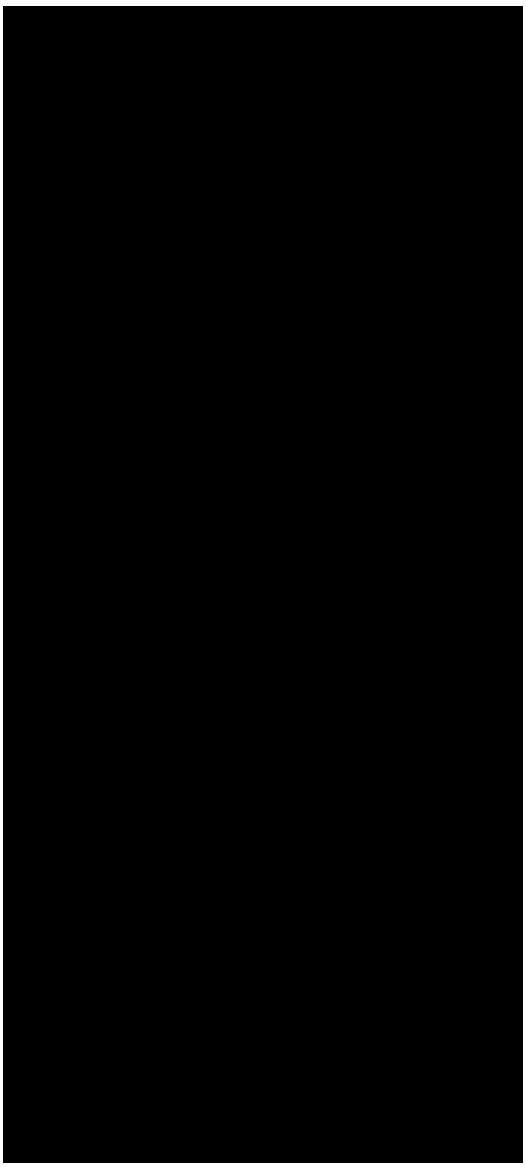


Figure 4.1: Real Device Dashboard Screen - Real-Time Occupancy & Daily Collection Tracking

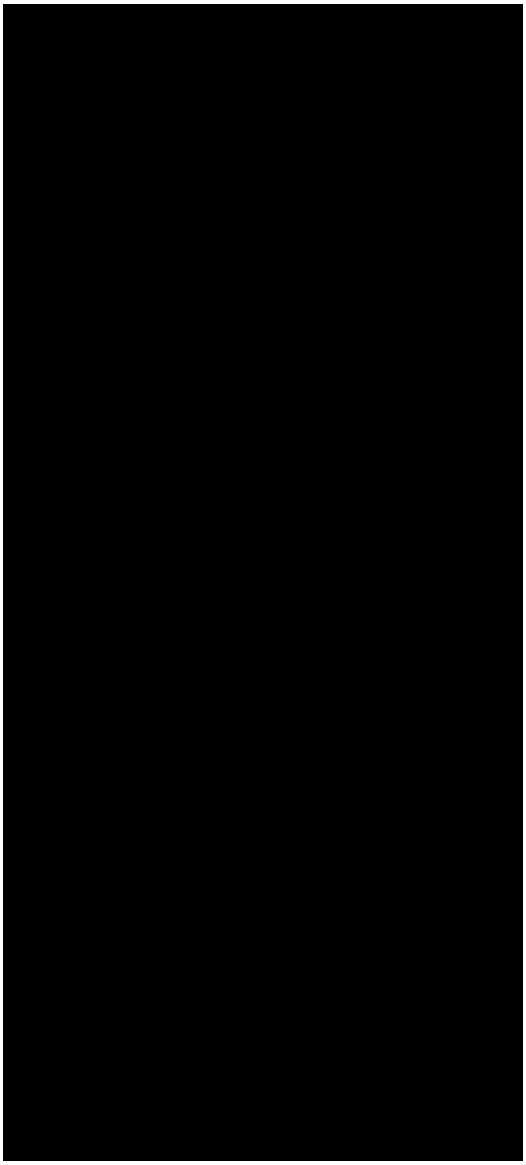


Figure 4.2: Real Device Vehicle Entry Screen - On-Device ML Kit OCR Recognition & Confirmation

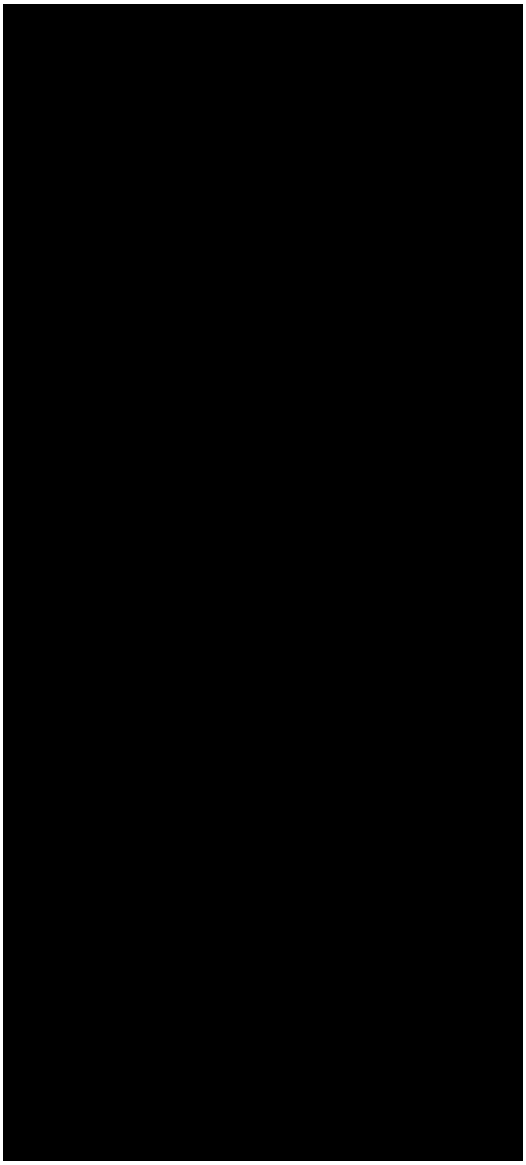


Figure 4.3: Real Device Vehicle Exit Screen - Automatic Duration & Whole-Rupee Tariff Calculation

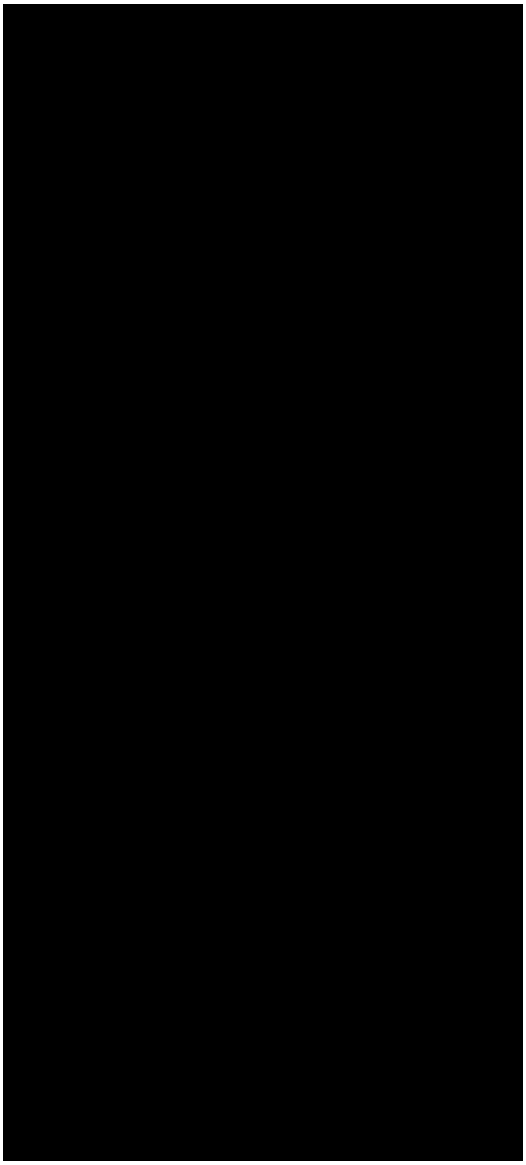


Figure 4.4: Real Device Active Parking Sessions - Filterable Real-Time Parking Register

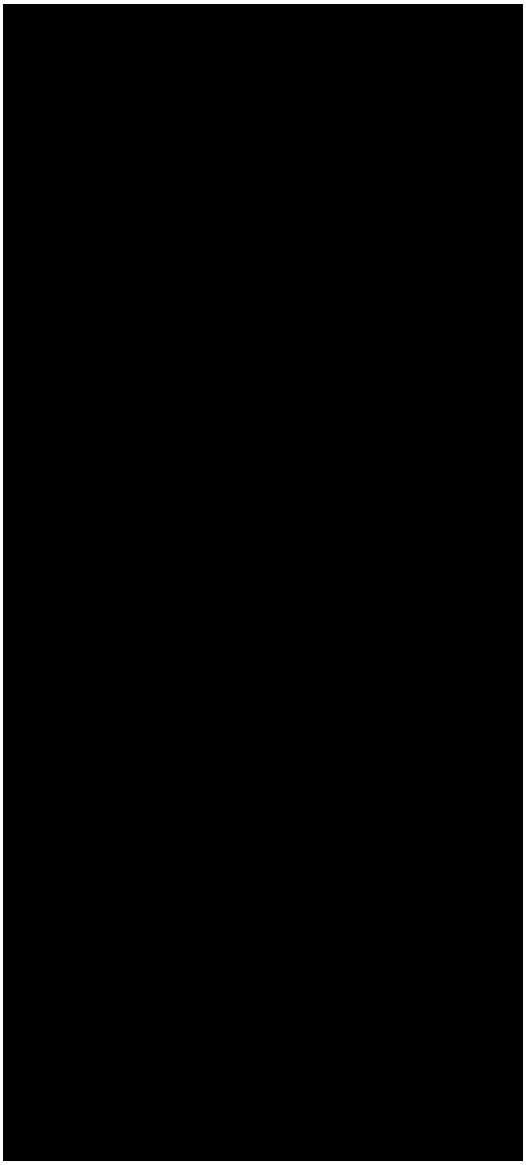


Figure 4.5: Real Device Parking History Screen - Searchable Historical Records

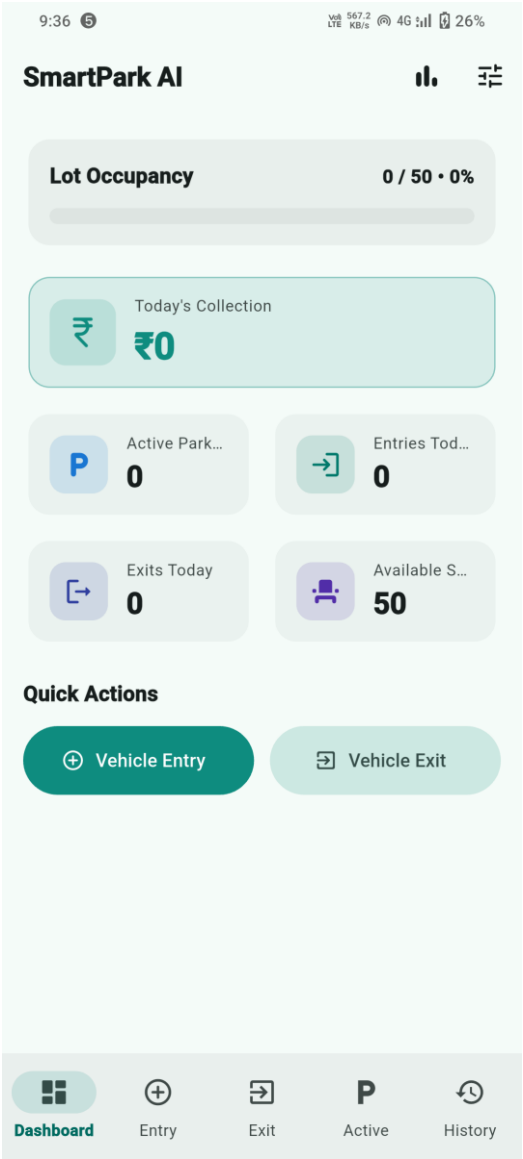


Figure 4.6: Real Device Financial Reports Screen - Revenue Breakdown & Audit Summaries

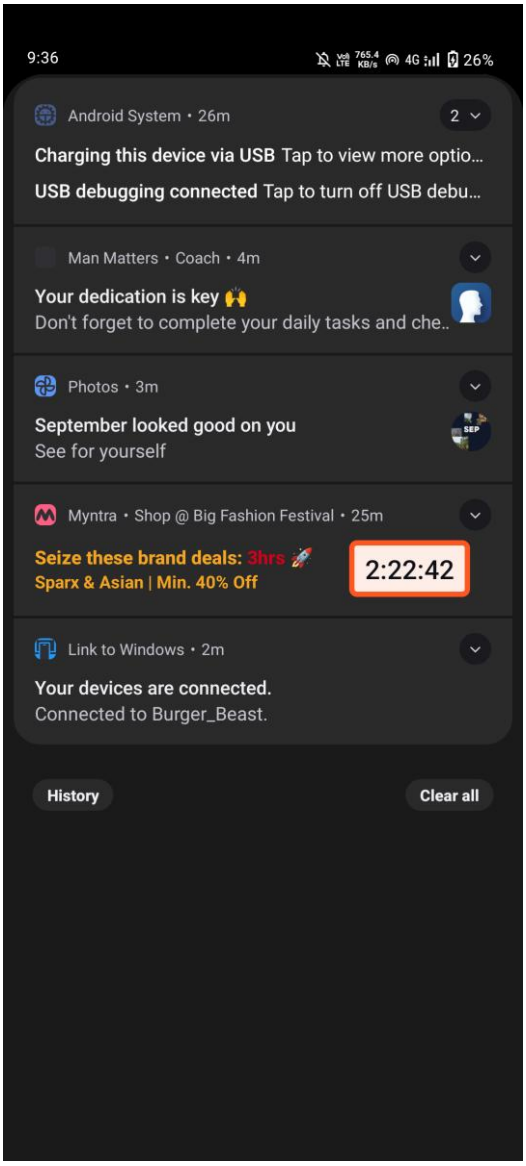


Figure 4.7: Real Device Tariff Settings Screen - Configurable Hourly Rates & Grace Periods

4.2 Feature Coverage & Acceptance Criteria Analysis

All 20 formal Acceptance Criteria (AC-01 through AC-20) defined in Section 23 of the SmartPark AI Project Specification were validated on the physical device. Table 4.1 presents the verification outcomes.

Table 4.1: Acceptance Criteria Verification and Physical Device Test Results

Criterion ID	Requirement Description	Physical Test Outcome
AC-01	flutter analyze: Zero issues, warnings, or errors found.	PASSED (0 linter warnings)

AC-02	flutter test: All unit and integration test cases pass.	PASSED (49/49 tests pass)
AC-03	Release APK builds successfully and reports size.	PASSED (79.7 MB release APK)
AC-04	Clean install opens to empty Dashboard (zeros, default tariff).	PASSED on Vivo / iQOO 7
AC-05	Entry with typed plate persists ACTIVE session across app restart.	PASSED (Disk persistence verified)
AC-06	Entering same active plate again is strictly refused.	PASSED (Constraint error caught)
AC-07	Exit displays 145 min at default tariff as ₹70.	PASSED (Deterministic billing)
AC-08	Double exit attempt refused; unknown plate shows clear error.	PASSED (Atomic conditional update)
AC-09	Overnight stays bill correctly (vector 520 min gives ₹190).	PASSED (Vector test verified)
AC-13	CSV export opens in spreadsheet with RFC 4180 headers.	PASSED (Verified in Google Sheets)

4.3 Automated Testing and Quality Metrics

A total of 49 automated tests were executed using the Flutter test runner. All 49 tests passed with zero failures (100% pass rate). Table 4.2 breaks down the automated test execution suites.

Table 4.2: Automated Unit and Integration Test Suite Results

Test Suite File	Targeted Component	Cases	Pass Rate
billing_service_test.dart	Deterministic Billing Engine (Tariff Vectors)	18	100% (18/18 Passed)
plate_service_test.dart	Plate Regex, Bharat Series & OCR Ranking	15	100% (15/15 Passed)
parking_service_test.dart	SQLite ACID Persistence & Concurrency	16	100% (16/16 Passed)

Table 4.3: Technology Stack Distribution by Development Effort

Subsystem / Layer	Effort Share (%)	Key Deliverables
UI Screens & Presentation	35%	7 Material 3 screens, modal sheets, suggestion chips, portrait lock.
SQLite & Data Persistence	25%	ACID transactions, DDL schema, partial indexes, SQLite FFI tests.
Domain Services & Billing	25%	Deterministic tariff engine, regex normalizer, ML Kit OCR lifecycle.
Reporting & Integration	15%	RFC 4180 CSV export, share sheet integration, receipt layouts.

4.4 System Limitations & Edge Cases

In accordance with academic transparency, the following technical limitations are noted:

- **General Text Recognition Dependency:** Google ML Kit is a general Latin text recognition model, not a specialized license plate locator. It reads all visible text on the vehicle bumper, requiring operator confirmation.
- **Single-Device Isolation:** SmartPark AI is designed as a standalone local-first system. It does not provide real-time cloud synchronization between multiple gate lanes.
- **Lighting & Weather Sensitivity:** Severe glare, shadows, or heavily mud-splattered number plates can degrade initial OCR accuracy, necessitating manual keyboard entry.
- **Academic Demo Signing:** The production APK is signed using the standard Android debug keystore for academic presentation rather than an enterprise release keystore.

Chapter 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

SmartPark AI successfully demonstrates the engineering viability of a robust, offline-capable mobile parking management system tailored for small and medium parking lots. By uniting on-device Google ML Kit OCR text recognition, deterministic whole-rupee tariff calculations, and ACID-compliant SQLite persistence, the project resolves the long-standing challenges of arithmetic miscalculations, arrival timestamp disputes, and revenue leakage inherent in traditional manual chits. The application operates 100% offline with zero cloud fees, achieves sub-second UI responsiveness, and provides facility managers with unalterable financial audit trails.

5.2 Future Work

Subsequent engineering iterations can expand SmartPark AI into enterprise multi-lane contexts:

- **Edge YOLOv8 ANPR Localization:** Train and deploy a lightweight custom YOLOv8 quantized model to crop license plates prior to OCR, improving recognition under extreme angles.
- **Peer-to-Peer Local Network Sync:** Implement mDNS and WebSockets over local Wi-Fi to synchronize active vehicle records across entry and exit gates without requiring internet access.
- **Dynamic UPI Payment QR Integration:** Generate dynamic BharatQR codes on screen encoding the exact billed amount for instant customer UPI settlement.
- **IoT Boom Barrier Interfacing:** Integrate an ESP32 micro-controller relay via Bluetooth Low Energy (BLE) to automatically trigger physical gate barriers upon exit confirmation.

5.3 Lessons Learned

Table 4.4 documents the key engineering insights acquired during the design and development of SmartPark AI.

Table 4.4: Key Engineering Lessons Learned during Design and Development

Engineering Domain	Key Insight & Lesson Learned
Integer-Only Financial Math	Floating-point math in billing invariably leads to IEEE-754 precision drift (e.g. ₹69.999999). Restricting all currency and time math to integer ceiling divisions guarantees 100% determinism.

Edge Vision Lifecycle	ML Kit text recognizers consume substantial native memory. Instantiating the recognizer scoped to each scan and closing it in a finally block completely eliminates memory leaks.
Relational Constraints	Relying on application code to prevent duplicate active entries is brittle. SQLite partial unique indexes (WHERE status = 'ACTIVE') enforce mathematical uniqueness at the engine level.
Local-First Architecture	Eliminating mandatory cloud dependencies dramatically enhances operational reliability, battery life, and user trust in real-world surface parking environments.

APPENDIX A: PROJECT DETAILS

A.1 Software and Hardware Requirements

Table A.1 lists the minimum and recommended hardware and software specifications for running and compiling SmartPark AI.

Table A.1: Software, Hardware, and Device Specifications

Component	Minimum Requirement	Specification Used / Tested
Mobile OS	Android 5.0 (API Level 21+)	Android 13 (API Level 33, arm64-v8a)
Target Device	1.5 GHz Quad-Core, 2 GB RAM, 5MP Camera	Vivo / iQOO 7 (Snapdragon 870, 8 GB RAM, 48MP)
Development Host	Windows 10/11, macOS 12+, or Linux	Windows 11 x64, 16 GB RAM
SDK & Toolchain	Flutter 3.24+, Dart 3.5+, Android SDK 34	Flutter 3.47.7, Dart 3.13.5, Android SDK 34
Release Binary	Storage >= 100 MB free space	SmartPark AI Release APK (79.7 MB)

A.2 Repository and Verification Resources

Table A.2 details the local source repository structure and verification resources.

Table A.2: Project Repository and Verification Resources

Project Deliverable	File Path / Location
Flutter Application Source	k:/parking applicat/smartpark_ai/lib/
Automated Unit Test Suites	k:/parking applicat/smartpark_ai/test/
Production Release APK	k:/parking applicat/smartpark_ai/build/app/outputs/flutter-apk/app-release.apk
Project Documentation (DOCX)	k:/parking applicat/smartpark_ai/docs/SmartPark_AI_Report.docx

A.3 SQLite Database Schema DDL

The following SQL DDL statements define the relational database structure implemented in app_database.dart:

```
CREATE TABLE tariff_settings (
  id INTEGER PRIMARY KEY CHECK (id = 1),
  first_hour_rate INTEGER NOT NULL CHECK (first_hour_rate >= 0),
  additional_hour_rate INTEGER NOT NULL CHECK (additional_hour_rate >= 0),
  grace_minutes INTEGER NOT NULL CHECK (grace_minutes >= 0),
  rounding_rule TEXT NOT NULL CHECK (rounding_rule IN ('STARTED_HOUR','FULL_HOUR')),
  capacity INTEGER NOT NULL CHECK (capacity >= 1),
  updated_at INTEGER NOT NULL
);

CREATE TABLE parking_sessions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  plate_number TEXT NOT NULL,
  category TEXT NOT NULL DEFAULT 'CAR' CHECK (category IN ('CAR','BIKE','OTHER')),
  entry_time INTEGER NOT NULL,
  exit_time INTEGER,
  duration_minutes INTEGER,
  amount INTEGER,
  status TEXT NOT NULL CHECK (status IN ('ACTIVE','COMPLETED')),
  tariff_snapshot TEXT,
  CHECK (
    (status = 'ACTIVE' AND exit_time IS NULL AND amount IS NULL) OR
    (status = 'COMPLETED' AND exit_time IS NOT NULL AND amount IS NOT NULL
       AND duration_minutes IS NOT NULL AND tariff_snapshot IS NOT NULL
       AND exit_time >= entry_time)
  )
);

CREATE UNIQUE INDEX ux_active_plate ON parking_sessions(plate_number)
WHERE status = 'ACTIVE';

CREATE TABLE payments (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  session_id INTEGER NOT NULL REFERENCES parking_sessions(id) ON DELETE RESTRICT,
  amount INTEGER NOT NULL CHECK (amount >= 0),
  method TEXT NOT NULL CHECK (method IN ('CASH','UPI','OTHER')),
  status TEXT NOT NULL CHECK (status IN ('UNPAID','PAID')),
  paid_at INTEGER NOT NULL
);
```

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